Factored Form of a Quadratic Function

Name _____

Algebra 2 Lesson 2-3 (fast track) Date _____ Period _____

Learning goal: I can solve a quadratic equation by factoring and write a quadratic function in factored form from its zeros and one more point.

Language goal: I can explain how the factors of a quadratic give its zeros using the words *factor*, *zero*, *Zero Product Property*, and *x-intercept*.

On the quiz: You will (1) factor a trinomial whose leading coefficient is not 1, using the grouping step from Example 3, Part B (bonus); (2) solve a quadratic equation such as $x^2+bx=c$ by factoring, as in Example 3; (3) find when a launched object lands (full Savvas quiz only); (4) find where a quadratic function is positive (full Savvas quiz only); and (5) write the equation of a parabola in a word problem from its two zeros and one more point, as in Example 6 and the Practice.

Zero Product Property: if $ab=0$, then $a=0$ or $b=0$ (or both). To use it, rewrite the equation so one side is 0, factor, then set each **factor** equal to 0. Spanish: *Propiedad del cero del producto*; zero of a function = *cero de una función*.

Example 3: Solve Quadratic Equations by Factoring

Solve the equation.

A. $x^2+x=42$

1. Get 0 on one side:
$x^2+x-42=0$

2. Factor. Two numbers with product -42 and sum 1: 7 and -6.
$(x+7)(x-6)=0$

3. Zero Product Property:
$x+7=0$ or $x-6=0$

4. $x=-7$ or $x=6$

Check: $(-7)^2+(-7)=49-7=42$ ✓
$6^2+6=36+6=42$ ✓

B. $2x^2=-9x+5$

1. Get 0 on one side:
$2x^2+9x-5=0$

2. $a\cdot c=2(-5)=-10$. Two factors of -10 whose sum is 9: -1 and 10. Rewrite $9x$ as $-x+10x$:
$2x^2-x+10x-5=0$

3. Factor by grouping:
$x(2x-1)+5(2x-1)=0$
$(x+5)(2x-1)=0$

4. Zero Product Property:
$x+5=0$ or $2x-1=0$

5. $x=-5$ or $x=\frac{1}{2}$

Check: $2(-5)^2=50$ and $-9(-5)+5=50$ ✓
$2\left(\frac{1}{2}\right)^2=\frac{1}{2}$ and $-9\left(\frac{1}{2}\right)+5=\frac{1}{2}$ ✓

Say it: In Part B, the factors are _____ and _____. By the *Zero Product Property*, one factor must be 0, so the *zeros* are $x=$ _____ and $x=$ _____. They are the *x-intercepts* of $y=2x^2+9x-5$ because ...

Try It 3: Solve the equation by factoring

Solve the equation by factoring.

Hint: get 0 on one side, factor, then use the Zero Product Property.

a. $x^2 + 8x = 20$

$x =$ _____ or $x =$ _____

b. $2x^2 = 3x + 2$

$x =$ _____ or $x =$ _____

Example 6: Write the Equation of a Parabola in Factored Form

Write an equation of a parabola with x-intercepts at $(-2, 0)$ and $(-1, 0)$ and which passes through the point $(-3, 20)$.

1. Factored form: $y = a(x - p)(x - q)$, $a \neq 0$, where p and q are the zeros.
2. Substitute the zeros $p = -2$, $q = -1$:
   $y = a(x - (-2))(x - (-1))$
   $y = a(x + 2)(x + 1)$
3. Substitute the point $(-3, 20)$ to find a:
   $20 = a(-3 + 2)(-3 + 1)$
   $20 = a(-1)(-2)$
   $20 = 2a$, so $a = 10$
4. $y = 10(x + 2)(x + 1)$

Watch the signs. If $x = -2$ is an x-intercept, the factor is $(x + 2)$, not $(x - 2)$: $x = -2$ is the number that makes $x + 2 = 0$.

Check: at $x = -3$,
$y = 10(-3 + 2)(-3 + 1) = 10(-1)(-2)$
$= 20$ ✓
At $x = -2$ and $x = -1$ one factor is 0, so $y = 0$ ✓

Why the extra point? Every $y = a(x + 2)(x + 1)$ with $a \neq 0$ has these two zeros. The point $(-3, 20)$ picks the one value of a.

Say it: The *x-intercepts* give the *zeros* -2 and -1, so the *factors* are _____ and _____ (by the *Zero Product Property*, $y = 0$ only when a factor is 0). I used the point $(-3, 20)$ to find $a =$ _____ because ...

Try It 6: Write the equation of a parabola

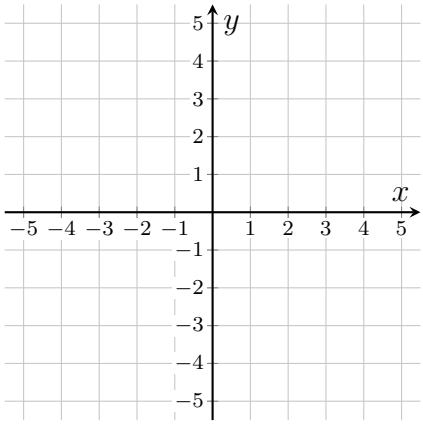
Write an equation of a parabola with x-intercepts at $(3, 0)$ and $(-3, 0)$ and which passes through the point $(1, 2)$.

Factored form with the zeros:

$y = a\,(_____)(_____)$

Substitute $(1, 2)$ and solve for a:

Plot the three points (optional):



Equation: $y =$ _____

Practice: Pumpkin launch (Performance Task 45)

At a state fair, a competition is held to launch pumpkins from catapults. A pumpkin is launched from the ground into the air and lands 4.5 s later. Its maximum height is 81 ft.

Part A Write a function that models the height, in feet, of the pumpkin x seconds after it is launched. Explain how you found the function.

Zeros: $x =$ _____ and $x =$ _____

The peak is at $x =$ _____ because ...

$y = a\,(_____)(_____)$

Point at the peak: (_____, _____)

Part B A second pumpkin is launched from the ground. After 1 second, it is 64 feet high. The pumpkin lands after 5 seconds. What is the maximum height of the pumpkin? Explain.

Zeros: $x =$ _____ and $x =$ _____ Point: (_____, _____) $y =$ _____

Maximum height: _____ ft at $x =$ _____ s

IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. Do the starred skills first. **SmartScore 70 or above on each starred (★) skill replaces your quiz score.**

Code	Skill	Feeds
TRU	Solve a quadratic equation using the zero product property	Warm-up for Example 3 (optional): the factors are already given
UB5 ★	Factor quadratics	Quiz 1 (bonus: factor $ax^2 + bx + c$) and the factoring step of Quiz 2
CJC ★	Solve a quadratic equation by factoring	Quiz 2: Example 3 and Try It 3
HPB ★	Write a quadratic function in factored form from its x-intercepts and another point	Quiz 5: Example 6, Try It 6, the Practice
G2Q	Write a quadratic function from its zeros	Warm-up for Example 6 (optional)

Optional, not on the short form: NMZ Factor out a monomial (the -16 step of Quiz 3, full Savvas quiz only; also the Practice); DP9 Graph solutions to quadratic inequalities (Quiz 4, full Savvas quiz only).

Before the quiz, finish these sentences.

To solve a quadratic equation by factoring, I first get _____ on one side of the equation.

By the *Zero Product Property*, if $(x + 4)(x - 2) = 0$, then $x =$ _____ or $x =$ _____.

Each *zero* of a quadratic function is an *x-intercept* of its graph because

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If a parabola has zeros p and q, its equation is $y =$ _____. I need one more point to find _____.

Exit question. A student says that the zeros of $y = (x - 2)(x + 7)$ are -2 and 7. Is the student correct? If not, describe and correct the error the student made.